

KRANTHI CHAITHANYA THOTA

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SUMMARY

Data Engineer with experience in designing and scaling cloud-native data platforms across GCP, with exposure in multicloud ecosystems. Proven expertise in data architecture (HLD/LLD), ETL/ELT pipelines, and GenAI-driven applications, alongside strong CI/CD automation practices. Demonstrated technical leadership through architectural ownership, cross-functional collaboration, mentorship to deliver reliable, production-ready data solutions.

CERTIFICATIONS

- Google Cloud Professional Data Engineer
- Google Cloud Associate Cloud Engineer
- HashiCorp Certified Terraform Associate

PROFESSIONAL EXPERIENCE

Environmental Finance Center (EFC) at University of North Carolina

Chapel Hill, NC

Data Engineer

Sep 2025 – Present

- Cut ML retraining turnaround by **55%** (5 hrs to 2.5 hrs) by optimizing a PySpark feature pipeline on AWS EMR, eliminating manual intervention in automated compute workflows.
- Saved analysts **500+ hours** annually by building an AI-powered table extraction pipeline that processed PDF documents into analysis-ready Excel outputs, eliminating manual data entry from the research workflow.
- Led CI/CD implementation for pipelines, SQL objects, and data models using Github Actions, improving deployment consistency.
- Introduced dynamic resource scaling in cloud infrastructure, reducing compute costs by approximately 20–30%.
- Reduced average query time by **60%** for downstream reporting teams by architecting a Star Schema data warehouse consolidating five federal datasets and implementing views for high-frequency access patterns.
- Developed organization-level proof-of-concepts (POCs) and contributed to the internal Community of Practice, driving standardization and best practices.
- Reviewed code, pull requests, and implementation strategies to ensure compliance with architectural standards, performance benchmarks, and security guidelines.

Clarkson CEM Group

Potsdam, NY

Data Engineer

May 2025 – Aug 2025

- Developed a modular data masking framework, enabling secure, reusable, and cross-team adoption of data protection standards.
- Designed pipeline architectures, achieving ~33% cost savings and ~50% reduction in execution time.
- Engineered memory-optimized Python ETL pipelines for legacy APIs, reducing ingestion time by 3× and improving system reliability.
- Migrated **20+ years** of IPEDS data into a central warehouse and built **30+** KPI dashboards for the Institutional Research team. Built a self-service **Power BI** forecasting platform used across **10+** departments, saving **150+** analyst hours annually.
- Provided technical support for solving complex engineering problems, acting as SME for Python, ML and cloud.
- Mentored engineers through technical direction and best-practice sharing, enabling skill development and certification readiness.

Egen (formerly SpringML)

Associate Data Engineer

Hyderabad, India

Jan 2022 – Dec 2023

- Built scalable data pipelines and workflows, improving data ingestion efficiency across production systems.
- Enabled EV charger site planning across New York State for **NY Power Authority** by engineering a geospatial pipeline processing **10+** public GIS datasets, delivering a map-based planning tool and location scoring API used by government stakeholders.
- Accelerated client analytics delivery by architecting a **Snowflake** warehouse using **dbt** with **40+** models across staging, intermediate, and mart layers shifting from hourly batch to near-real-time updates via **CDC**.
- Eliminated a manual patent data extracted workflow by building a **Document AI** pipeline ingesting patent documents via **Cloud Run**, feeding live client dashboards in **BigQuery**.
- Reduced pipeline runtime by **33%** (6 hrs to 4 hrs) and lowered production failure rates by refactoring multi-stage **Airflow** DAGs into dynamic task generation.
- Automated GCP environment provisioning for **50+** resources via **Terraform** and Cloud Build, cutting setup turnaround by **87%** (4 hrs to 30 min) with audit-ready IAM and security guardrails.

PROJECTS

Real-Time Reddit Stock Sentiment Tracker | *Python, TimescaleDB, Reddit API*

- Built a streaming pipeline processing **100+** Reddit comments per second, surfacing breakout stock tickers within 5 seconds using tumbling window logic and TimescaleDB for degradation-free write throughput.

HAVK Mladost Sports Club Data Infrastructure | *PostgreSQL, Python, Looker Studio*

- Replaced Excel-based workflows with a normalized PostgreSQL schema and automated Python ETL pipeline; delivered Looker Studio dashboards tracking membership, revenue, and churn risk, improving renewal visibility for club leadership.

Serverless Data Job Deployment Framework | *Terraform, GCP, GitHub Actions*

- Reduced GCP data job deployment time from **3+ hrs to under 10 minutes** across 3 teams by building a reusable Terraform framework with standardized IAM roles, secret management, and logging modules, eliminating manual configuration errors.

SKILLS

Programming & Scripting: Python, SQL, PySpark, JavaScript, PowerShell, Bash

Cloud & Infrastructure: GCP (BigQuery, Dataflow, Cloud Run, Pub/Sub, Vertex AI), AWS (EMR, Redshift, Glue, Lambda, S3), Terraform, Docker, Kubernetes

Data Engineering: Data Warehousing, Kafka, Delta Lake, Pub/Sub, ETL/ELT Development, Data Modeling, API Integration, Data Masking

GenAI & LLMs: PyTorch, Transformers (BERT, LLMs, RAG), LangChain, OpenClaw

DevOps & Version Control: Git, GitHub, Azure DevOps, CI/CD Pipelines

GIS & Databases: ArcGIS Pro, QGIS, PostGIS, PostgreSQL, MongoDB, SqlServer, MySQL, TimescaleDB

EDUCATION

Clarkson University

Master of Science – Applied Data Science; CGPA: 4.0/4.0

Potsdam, USA

Jan 2024 – Aug 2025

Kakatiya Institute of Technology and Science

Bachelor of Technology – Electronics and Communication Engineering; CGPA: 3.54/4.0

Warangal, India

Aug 2018 – May 2022